

CL papers AI agent

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Demo Outputs

```
mode,retrieval_weight,graph_weight,quality_weight,use_graph,num_final_chunks,vector_chunks
vector_only,1.0,0.0,0.0,False,5,5,0,0,3,0,1.0,True,144.49,0.0,144.49
graph_guided,0.5,0.4,0.1,True,8,1,7,0,5,5,1.0,True,154.99,83.5,239.49
hybrid_graphrag,0.8,0.1,0.1,True,8,5,3,0,5,5,1.0,True,99.5,99.7,199.2
```

```
mode,num_seed_chunks,num_graph_papers,num_candidate_chunks,num_final_chunks,citation_cover
vector_only,5,0,5,5,1.0,True,157.0,0.0,0.0,0.0,157.0,C:\Users\97150\Documents\GitHub\CLPapers
graph_guided,5,5,15,8,1.0,True,156.48,88.5,0.0,0.0,244.98,C:\Users\97150\Documents\GitHub\CLP
hybrid_graphrag,5,5,15,8,1.0,True,102.07,86.11,0.0,0.0,188.18,C:\Users\97150\Documents\GitHub
```

Figure1 . Sample comparison of Vector-Only, Graph-Guided, and Hybrid GraphRAG retrieval outputs.

Figure shows a comparison of the three retrieval setups that were considered in the final version of the GraphRAG system: Vector-Only, Graph-Guided, and Hybrid GraphRAG. Vector-Only retrieves documents only via semantic search, whereas Graph-Guided extends retrieved papers via Neo4j graph relationships. Hybrid GraphRAG retrieval setup uses a combination of semantic search and graph search to extend and rank retrieved papers. Figure demonstrates that graph-augmented approaches are capable of retrieving more graph-relevant documents and candidate chunks. Hybrid GraphRAG setup achieves a compromise between retrieval accuracy and graph reasoning using semantic and graph connections.

Safety Mitigation

```
1 question_id,mode,rank,paper_id,title,chunk_id,pages,source,graph_reasons
2 q_api_001,vector_only,1,paper85,Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented Generati
3 q_api_001,vector_only,2,paper148,H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Conversations
4 q_api_001,vector_only,3,paper148,H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Conversations
5 q_api_001,vector_only,4,paper12,Generation: A Controlled Empirical Study1,paper12_chunk_0000,1-2,vector,[]
6 q_api_001,vector_only,5,paper12,Generation: A Controlled Empirical Study1,paper12_chunk_0017,14,vector,[]
7 q_api_001,graph_guided,1,paper70,Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning,paper70_chunk_001
8 q_api_001,graph_guided,2,paper63,FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoning,paper6
9 q_api_001,graph_guided,3,paper70,Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning,paper70_chunk_000
10 q_api_001,graph_guided,4,paper63,FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoning,paper6
11 q_api_001,graph_guided,5,paper111,GRAIL: A Deep-Granularity Hybrid Resonance Framework for Real-Time Agent Discovery via SLM-Enhanc
12 q_api_001,graph_guided,6,paper85,Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented Generat
13 q_api_001,graph_guided,7,paper121,LLM-Augmented Semantic Steering of Text Embedding Projection Spaces,paper121_chunk_0013,6-7,graph
14 q_api_001,graph_guided,8,paper111,GRAIL: A Deep-Granularity Hybrid Resonance Framework for Real-Time Agent Discovery via SLM-Enhanc
15 q_api_001,hybrid_graphrag,1,paper85,Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented Gener
16 q_api_001,hybrid_graphrag,2,paper70,Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning,paper70_chunk_
17 q_api_001,hybrid_graphrag,3,paper148,H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Conversat
18 q_api_001,hybrid_graphrag,4,paper63,FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoning,pap
19 q_api_001,hybrid_graphrag,5,paper148,H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Conversat
20 q_api_001,hybrid_graphrag,6,paper12,Generation: A Controlled Empirical Study1,paper12_chunk_0000,1-2,vector,[]
21 q_api_001,hybrid_graphrag,7,paper12,Generation: A Controlled Empirical Study1,paper12_chunk_0017,14,vector,[]
22 q_api_001,hybrid_graphrag,8,paper70,Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning,paper70_chunk_
23 q_api_002,vector_only,1,paper12,Generation: A Controlled Empirical Study1,paper12_chunk_0017,14,vector,[]
24 q_api_002,vector_only,2,paper12,Generation: A Controlled Empirical Study1,paper12_chunk_0000,1-2,vector,[]
25 q_api_002,vector_only,3,paper148,H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Conversations
26 q_api_002,vector_only,4,paper85,Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented Generati
27 q_api_002,vector_only,5,paper85,Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented Generati
28 q_api_002,graph_guided,1,paper70,Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning,paper70_chunk_000
29 q_api_002,graph_guided,2,paper63,FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoning,paper6
30 q_api_002,graph_guided,3,paper70,Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning,paper70_chunk_000
31 q_api_002,graph_guided,4,paper63,FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoning,paper6
32 q_api_002,graph_guided,5,paper79,"A Multi-View Media Profiling Suite: Resources, Evaluation, and Analysis",paper79_chunk_0024,13,gr

1 question_id,mode,rank,source,paper_id,title,pages,retrieval_score,graph_score,final_score,graph_reasons
2 q_api_001,vector_only,1,vector,paper85,Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented G
3 q_api_001,vector_only,2,vector,paper148,H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Convers
4 q_api_001,vector_only,3,vector,paper148,H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Convers
5 q_api_001,vector_only,4,vector,paper12,Generation: A Controlled Empirical Study1,1-2,0.11574279345485947,0.0,0.1157,[]
6 q_api_001,vector_only,5,vector,paper12,Generation: A Controlled Empirical Study1,14,0.0,0.0,0.0,[]
7 q_api_001,graph_guided,1,graph,paper70,Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning,10,0.823265
8 q_api_001,graph_guided,2,graph,paper63,FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoning,1
9 q_api_001,graph_guided,3,graph,paper70,Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning,1,0.7395312
10 q_api_001,graph_guided,4,graph,paper63,FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoning,
11 q_api_001,graph_guided,5,graph,paper111,GRAIL: A Deep-Granularity Hybrid Resonance Framework for Real-Time Agent Discovery via SLM-I
12 q_api_001,graph_guided,6,vector,paper85,Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented
13 q_api_001,graph_guided,7,graph,paper121,LLM-Augmented Semantic Steering of Text Embedding Projection Spaces,6-7,0.38845807808738214
14 q_api_001,graph_guided,8,graph,paper111,GRAIL: A Deep-Granularity Hybrid Resonance Framework for Real-Time Agent Discovery via SLM-I
15 q_api_001,hybrid_graphrag,1,vector,paper85,Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augment
16 q_api_001,hybrid_graphrag,2,graph,paper70,Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning,10,0.823
17 q_api_001,hybrid_graphrag,3,vector,paper148,H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Co
18 q_api_001,hybrid_graphrag,4,graph,paper63,FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoni
19 q_api_001,hybrid_graphrag,5,vector,paper148,H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Co
20 q_api_001,hybrid_graphrag,6,vector,paper12,Generation: A Controlled Empirical Study1,1-2,0.8912892354718298,0.0,0.813,[]
21 q_api_001,hybrid_graphrag,7,vector,paper12,Generation: A Controlled Empirical Study1,14,0.8770597924184171,0.0,0.8016,[]
22 q_api_001,hybrid_graphrag,8,graph,paper70,Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning,1,0.7395
23 q_api_002,vector_only,1,vector,paper12,Generation: A Controlled Empirical Study1,14,1.0,0.0,1.0,[]
24 q_api_002,vector_only,2,vector,paper12,Generation: A Controlled Empirical Study1,1-2,0.533776200152738,0.0,0.5338,[]
25 q_api_002,vector_only,3,vector,paper148,H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Convers
26 q_api_002,vector_only,4,vector,paper85,Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented G
27 q_api_002,vector_only,5,vector,paper85,Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented G
28 q_api_002,graph_guided,1,graph,paper70,Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning,2-3,0.73449
29 q_api_002,graph_guided,2,graph,paper63,FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoning,
30 q_api_002,graph_guided,3,graph,paper70,Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning,3,0.6226434
31 q_api_002,graph_guided,4,graph,paper63,FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoning,
32 q_api_002,graph_guided,5,graph,paper79,"A Multi-View Media Profiling Suite: Resources, Evaluation, and Analysis",13,0.2318232657432
```

Figure 2. Evidence provenance and citation records generated during GraphRAG retrieval.

To increase answer reliability and avoid responses without any basis, the last GraphRAG system includes provenance and evidence-based retrieval techniques. For each retrieved chunk, its source paper, range, way of retrieval, and graph reasoning information is saved before its use in answer generation.

Figure X illustrates the evidence information stored during the retrieval procedure. All citations, sources, scores, and graphs are stored for each candidate chunks. In this way, it becomes possible to trace all generated answers to the original research papers that were used as evidence.

With the help of citation information and provenance of the retrieval procedure, the system is able to ensure that all answers will be related to the retrieved documents only and not generated by language models.

The provenance of the retrieval also provides an opportunity to find out what papers and chunks were used for particular purposes.

	parent_topic	confidence	hop_distance	topic	paper_id	hop_type
0	Large Language Models	0.95	2	Reasoning	paper15	sibling_topic
1	Large Language Models	0.82	2	Reasoning	paper21	sibling_topic
2	Large Language Models	0.82	2	Reasoning	paper23	sibling_topic
3	Large Language Models	0.82	2	Reasoning	paper24	sibling_topic
4	Large Language Models	0.82	2	Reasoning	paper27	sibling_topic
...	...	...	...	...	...	...
205	Large Language Models	0.82	2	Code Generation	paper24	sibling_topic
206	Large Language Models	0.82	2	Code Generation	paper34	sibling_topic
207	Large Language Models	0.82	2	Code Generation	paper78	sibling_topic
208	Large Language Models	0.82	2	Code Generation	paper126	sibling_topic
209	Large Language Models	0.82	2	Code Generation	paper135	sibling_topic

210 rows × 6 columns

Figure 3. Multi-hop topic expansion through the Neo4j topic hierarchy.

While D3 builds on the D2 method of retrieving papers using graphs, it does so by adding the concept of multi-hop topic expansion. In addition to searching for exact topic matches, the system uses SUBTOPIC_OF relations to traverse the topic hierarchy, thereby finding sibling topics that have the same parent topic. As can be seen from Figure X, for instance, papers related to the topics of

Reasoning and Code Generation were found using the parent topic of "Large Language Models".

	paper_id	title	graph_score	path_count	reasons
0	paper34	Enhanced LLM Reasoning by Optimizing Reward Fu...	3.475	6	[shared_topic: Code Generation, shared_topic: ...
1	paper24	MolViBench: Evaluating LLMs on Molecular Vibe ...	2.870	5	[shared_topic: Code Generation, shared_topic: ...
2	paper127	SciResearcher: Scaling Deep Research Agents fo...	2.655	5	[shared_topic: Reasoning, sibling_topic: Reaso...
3	paper43	Maistros: A Greek Large Language Model Adapted...	2.245	4	[shared_topic: Reasoning, sibling_topic: Reaso...
4	paper165	Do Reasoning Vision-Language Models Inversely ...	2.245	4	[shared_topic: Reasoning, sibling_topic: Reaso...
5	paper68	Medmarks: A Comprehensive Open-Source LLM Benc...	2.115	4	[shared_topic: Reasoning, sibling_topic: Reaso...
6	paper47	RMGAP: Benchmarking the Generalization of Rewa...	2.115	4	[shared_topic: Reasoning, sibling_topic: Reaso...
7	paper126	Feedback-Normalized Developer Memory for Reinf...	2.115	4	[shared_topic: Code Generation, sibling_topic:...

Figure 4. Ranked candidate papers after graph scoring and subgraph selection.

D3 introduces graph-aware candidate scoring and subgraph selection. Candidate papers are ranked according to multiple graph signals, including shared topics and author relationships. The highest-scoring papers are retained to construct a focused retrieval subgraph.

	version	topic_candidates	author_candidates	scored_subgraph	multi_path_aggregation
0	D2 (single-hop, unscored)	48	0	no	no
1	D3 (multi-hop, scored)	direct + sibling (hierarchy)	direct + 2-hop co-author network	yes (path_score, ranked, LIMIT)	yes (20 candidates after selection)

Table 1. Comparison between D2 and D3 graph retrieval approaches.

Compared with D2, the D3 implementation introduces multi-hop graph expansion, candidate scoring, evidence aggregation, and subgraph selection. These additions allow the system to prioritize stronger graph evidence and

Evaluation and Ablation Study

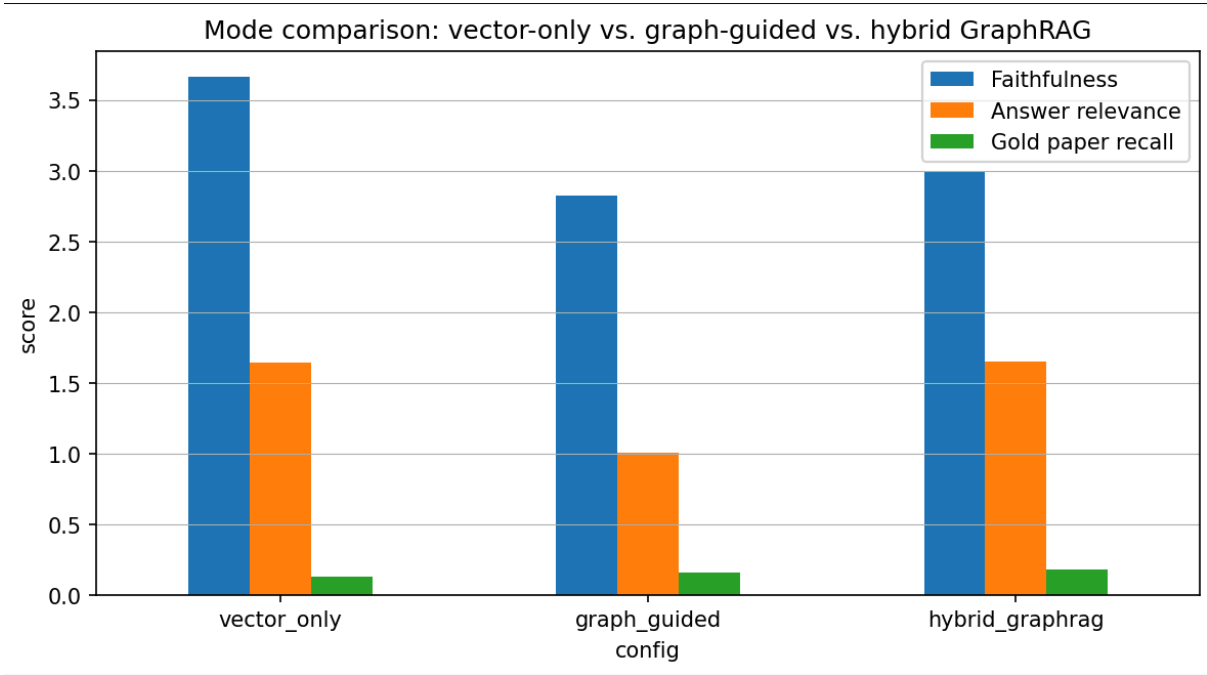


Figure 5. Performance comparison of Vector-Only, Graph-Guided, and Hybrid GraphRAG using faithfulness, answer relevance, and gold paper recall.

The three retrieval configurations were evaluated using faithfulness, answer relevance, and gold paper recall metrics. The comparison highlights the impact of graph-based retrieval on answer quality and retrieval effectiveness.

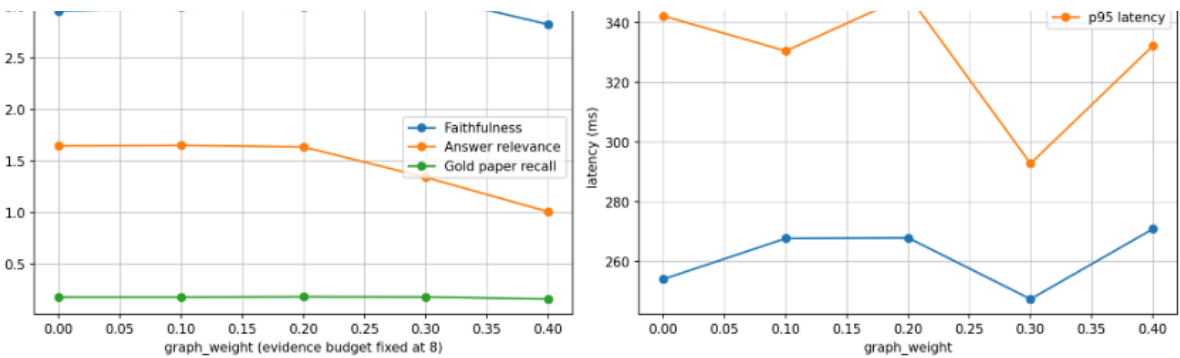


Figure 5 . Impact of graph weight on retrieval quality and latency.

Different graph weights were evaluated to analyze the trade-off between retrieval quality and system latency. The experiment helps identify a suitable balance between semantic retrieval and graph-based reasoning.

config,	retrieval_weight,	graph_weight,	quality_weight,	mean_faithfulness,	mean_answer_relevance,	mean_citation_coverage,	mean_gold_precision,
vector_only,	1.0,0.0,0.0,3.6641666666666666,	1.6450000000000002,	1.0,0.5777777777777778,	0.13261643449143448,	178.27583333333334,	229.77777777777777,	0.8166666666666667,
graph_guided,	0.5,0.4,0.1,2.8216666666666668,	1.0075,1.0,0.4180555555555557,	0.16116441428941428,	275.65916666666664,	334.955		
hybrid_graphrag,	0.8,0.1,0.1,2.9941666666666666,	1.6500000000000001,	1.0,0.46309523809523806,	0.18041192603692602,	257.64583333333333,	317	
grid_r1.g0.0.q0.0.1.0.0.0.0.2.9499999999999997,	1.6450000000000002,	1.0,0.46309523809523806,	0.18041192603692602,	254.11666666666666,			
grid_r0.8.g0.1.q0.1.0.8.0.1.0.1.2.9941666666666666,	1.6500000000000001,	1.0,0.46309523809523806,	0.18041192603692602,	267.7575,	330.5485		
grid_r0.7.g0.2.q0.1.0.7.0.2.0.1.2.99,1.635,1.0,0.4853174603174603,	0.18388414825914826,	267.94583333333333,	349.12249999999995				
grid_r0.6.g0.3.q0.1.0.6.0.3.0.1.3.1025000000000005,	1.34,1.0,0.48948412698412697,	0.18174739612239613,	247.43999999999997,	292.847			
grid_r0.5.g0.4.q0.1.0.5.0.4.0.1.2.8216666666666668,	1.0075,1.0,0.4180555555555557,	0.16116441428941428,	270.92416666666667,	332.2605			

Table 2 . Summary of evaluation metrics across the tested retrieval configurations.

The results show that Hybrid GraphRAG achieved the best overall balance between retrieval quality and graph reasoning. By combining semantic retrieval with graph-based expansion, it improved recall while maintaining strong answer relevance and faithfulness.

GraphRAG Integration

question_id	mode	rank	source	paper_id	title	pages	retrieval_score	graph_score	final_score	graph_reasons
q_api_001	vector_only	1	vector	paper85	Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented Generation	14	0.8	0.0	0.8	
q_api_001	vector_only	2	vector	paper148	H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Conversations	14	0.7	0.0	0.7	
q_api_001	vector_only	3	vector	paper148	H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Conversations	14	0.6	0.0	0.6	
q_api_001	vector_only	4	vector	paper12	Generation: A Controlled Empirical Study	11	0.1	0.0	0.1	
q_api_001	vector_only	5	vector	paper12	Generation: A Controlled Empirical Study	11	0.0	0.0	0.0	
q_api_001	graph_guided	1	graph	paper70	Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning	10	0.8	0.2	0.823265	
q_api_001	graph_guided	2	graph	paper63	FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoning	14	0.7	0.3	0.7395312	
q_api_001	graph_guided	3	graph	paper70	Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning	10	0.7	0.3	0.7395312	
q_api_001	graph_guided	4	graph	paper63	FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoning	14	0.6	0.4	0.64	
q_api_001	graph_guided	5	graph	paper111	GRAIL: A Deep-Granularity Hybrid Resonance Framework for Real-Time Agent Discovery via SLM-Inspired Retrieval-Augmented Generation	14	0.5	0.5	0.5	
q_api_001	graph_guided	6	vector	paper85	Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented Generation	14	0.4	0.6	0.46	
q_api_001	graph_guided	7	graph	paper121	LLM-Augmented Semantic Steering of Text Embedding Projection Spaces	6	0.3	0.8	0.38845807808738214	
q_api_001	graph_guided	8	graph	paper111	GRAIL: A Deep-Granularity Hybrid Resonance Framework for Real-Time Agent Discovery via SLM-Inspired Retrieval-Augmented Generation	14	0.3	0.8	0.38845807808738214	
q_api_001	hybrid_graphrag	1	vector	paper85	Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented Generation	14	0.8	0.2	0.823265	
q_api_001	hybrid_graphrag	2	graph	paper70	Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning	10	0.8	0.2	0.823265	
q_api_001	hybrid_graphrag	3	vector	paper148	H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Conversations	14	0.7	0.3	0.7395312	
q_api_001	hybrid_graphrag	4	graph	paper63	FT-RAG: A Fine-grained Retrieval-Augmented Generation Framework for Complex Table Reasoning	14	0.6	0.4	0.64	
q_api_001	hybrid_graphrag	5	vector	paper148	H-RAG at SemEval-2026 Task 8: Hierarchical Parent-Child Retrieval for Multi-Turn RAG Conversations	14	0.5	0.5	0.5	
q_api_001	hybrid_graphrag	6	vector	paper12	Generation: A Controlled Empirical Study	11	0.1	0.0	0.1	
q_api_001	hybrid_graphrag	7	vector	paper12	Generation: A Controlled Empirical Study	11	0.0	0.0	0.0	
q_api_001	hybrid_graphrag	8	graph	paper70	Verbal-R3: Verbal Reranker as the Missing Bridge between Retrieval and Reasoning	10	0.7	0.3	0.7395312	
q_api_002	vector_only	1	vector	paper12	Generation: A Controlled Empirical Study	11	0.1	0.0	0.1	
q_api_002	vector_only	2	vector	paper12	Generation: A Controlled Empirical Study	11	0.0	0.0	0.0	

Figure6. Evidence Provenance and Retrieval Records for GraphRAG Queries

Figure 6 presents a sample of the retrieval records generated by the GraphRAG system. For each query, the system stores the retrieved paper, retrieval method, ranking information, retrieval score, graph score, and graph reasoning evidence. These records provide provenance tracking and allow generated answers to be traced back to the original research papers, improving transparency and reliability.

